Architectural Styles

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There are different ways to view the organization of distributed systems:

- Software Architecture: The logical organization.
- System Architecture: The actual physical realization.

0.1 Component

A modular unit with well-defined interfaces that is replaceable within its environment.

0.2 Connector

A mechanism that mediates communication, coordination, or cooperation among components.

Examples:

- RPC (Remote Procedure Call)
- Messaging
- Streaming

0.2.1 Event-Based Architectures

Components communicate via events using the Publish/Subscribe paradigm.

- Components can publish and subscribe to events.
- Events are delivered to subscribers.
- Processes are decoupled in space.
- Processes are coupled in time.

Types:

- Subject-Based: A message belongs to groups, channels, or topics. Users subscribe to these groups and receive associated messages.
- Content-Based: Message delivery is based on a query or predicate issued by the subscriber.

0.2.2 Object-Based Architectures

- Components interact via RPC.
- Each object corresponds to a software component.

A remote object provides a service comprising a set of functions.
The service description includes function names, argument numbers and types, and return types.

CORBA Common Object Request Broker Architecture.
An industry standard for distributed objects programming.

Java RMI Remote Method Invocation allows an object running in one Java Virtual Machine to invoke methods on an object running in another Java VM.

Amazon S3 Objects (files) are placed into buckets (directories).
Buckets cannot be placed inside other buckets.

- PUT: Create a bucket or object.
- GET: List objects in a bucket.
- GET: Read an object using its URI.

1 What Are the Different Software Architectures?

- Layered Architectures
- Event-Based Architectures
- Data-Centered Architectures
- Object-Based Architectures
- Shared Data-Space Architectures

2 System Architectures

- Centralized: Client-Server systems and Multitiered systems (Servers as clients).
- Decentralized: Peer-to-Peer systems.
- Hybrid: Client-Server + Decentralized.

2.1 Layered Architecture

Components are organized in a layered fashion.

A component at layer L_i is allowed to call components at the lower layer L_{i-1}, not the other way around.

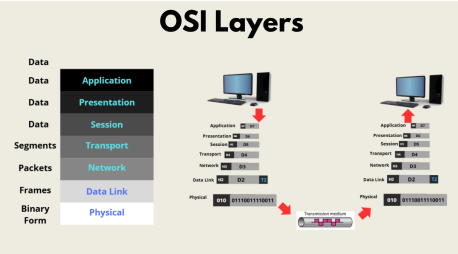
Control flows from layer to layer (down), and responses flow upwards (up).

2.1.1 Example: OSI Model

The OSI model is a conceptual model that characterizes and standardizes communication functions.

It consists of seven layers:

1. Application
2. Presentation
3. Session
4. Transport
5. Network
6. Data Link
7. Physical



2.1.2 Levels

- User-Interface Level: Provides access to the user with the system.
- Processing Level: Contains the applications in the system.
- Data Level: Manages the data in the system.

2.2 Data-Centered Architecture

Processes communicate through a common repository.

- Evolves around the data.
- Communication takes place through files.

- Examples: Web-based distributed systems and Distributed Database Systems.

2.3 Shared Data-Space Architecture

Combines Event-Based and Data-Centric architectural styles.
Processes communicate through a common repository.

- Publishers post items to a shared data space.
- Subscribers access items at a later time.
- Processes are decoupled in both space and time.

2.3.1 JavaSpaces

A JavaSpace is a Jini service that stores Java objects in memory.
It supports collaboration among services.
Operations:

- Write: Write an object to JavaSpace.
- Read: Return a copy and leave the object in JavaSpace.
- Take: Return a copy and remove the object from JavaSpace.
- Notify: Register a callback to be notified when changes occur.

2.4 Client-Server Architecture

- Client: A process that requests a service from a server.
- Server: A process that offers a service, such as a file or database service.

2.4.1 Steps at Client

1. Make a message for the server.
2. Identify the service the client wants.
3. Send the message to the server.

2.4.2 Steps at Server

1. Wait for an incoming request.
2. Process the request.
3. Package the result in a reply message.
4. Send the reply to the client.

Communication between client and server can use:

- Connectionless Protocol (UDP): Used when the underlying network is reliable, according to the slides.
- Connection-Oriented Protocol (TCP): Used when the underlying network is unreliable, according to the slides.

2.4.3 UDP Problems

- Occasional transmission failures may occur.
- The client resends the request when no reply arrives.
- The client cannot determine whether the request or reply was lost.
- Resending a request may execute the operation twice.

2.4.4 TCP Steps

1. The client establishes a connection with the server.
2. The client sends the request.
3. The server sends the result using the same connection.
4. The connection is closed after receiving the result.

Disadvantages mentioned in the slides:

- Low performance.
- Connection setup is costly for small messages.

2.5 Multitiered Architecture

Application programs are distributed across different machines.

2.5.1 Two-Tier Organization

- Client Machine: Contains programs implementing the user-interface level.
- Server Machine: Contains the processing and data levels.

2.5.2 Three-Tier Organization

- User Interface: Presentation level.
- Application Server: Processing level.
- Database Server: Data level.

Request flow:

1. User interface requests an operation.
2. Application server requests data.
3. Database server returns data.
4. Application server returns the result.

2.6 Layers vs. Tiers

- Layer: Logical layer, a term in Software Architecture.
- Tier: Physical layer, a term in System Architecture.

2.7 P2P Systems

Peer-to-Peer networks use distribution between equivalent peers.

This is called horizontal distribution, as opposed to layered vertical distribution.

Problems:

- Who should store a particular data item?
- How is a message routed through the peer group?
- Sending messages to all peers may flood the network.

2.8 Overlay Network

An overlay network is a logical network built on top of a physical network.

Focus at the application level.

2.9 Hybrid Architecture

Hybrid Architecture combines Decentralized and Client-Server systems.

2.10 Edge-Server Systems

Servers are placed at the edge of the network.

They are located at the boundary between enterprise networks and the Internet, such as an ISP.

2.11 Collaborative Distributed Systems

Components of a collaborative content distribution network:

- A component that redirects client requests to other servers.
- A component for analyzing access patterns.
- A component for managing the replication of Web pages.